using your own two hands has become a lot easier than before. But before that, let's clarify a few basic things first.

You can't take 3D prints of just about everything

A 3D printer is not some kind of a magic machine. You might discover an extremely attractive picture of an object with a complex assembly and multiple colors and figure it would be cool to get the real deal using your 3D printer.

But these are not easy to convert into real objects and usually require special modifications to be made in your regular printer.

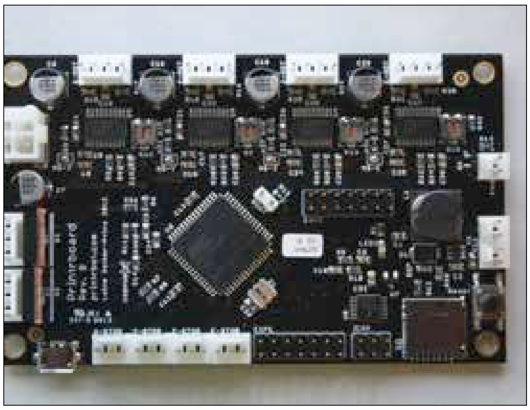
What most 3D printers can definitely create are moderately-sized objects in single colors and simple structures. For instance, I got into 3D printing because I wanted individual parts of a device for my project. And of course, for the cool factor!

You need to learn 3D modeling if you want to fully utilize your 3D printer

Sure, there are several useful websites like Thingiverse and Youmagine, which provide rookies with thousands of 3D models that they can print, as well as a few like Myminifactory which direct you to only those files with physical features that can be effortlessly converted into 3D prints, on the specific device that you are dealing with.

But if that's the only function you want from your 3D printer, then it's no better than an expensive toy. Learn 3D modeling, however, and it'll open up a whole new world of possibilities – you can then make individual parts of your own and assemble them to build complex machinery.

You can even fashion instruments to use around your house.



One of the many components that you will need

Keep it simple at first

Don't go for a fancy dual extruder or a 3D printer with a very large build volume from the word go. Keep it simple in the beginning. Build your own 3D printer and learn from it by tinkering around. You learn a lot more from building machines than buying them.

They aren't plug and play

3D printers are not foolproof; none of the models made so far have achieved a 100 per cent success rate. Why, even your basic paper printers have jam issues. The performance of 3D printers, in terms of speed, reliability and detailing, varies from machine to machine.

So be prepared for some minor tinkering around when you build one of your own. Since you've put it together from scratch, you know the ins and outs of it, and it's therefore easier to fix.

It's a learning process

3D printers are like musical instruments, you will always keep discovering new facets every time you use them. It's a fast-evolving field, with technological updates pouring in after every few months.

Plus, there's ample help and technical support available from the 3D printers' online community, which provides useful tips on upgrades and maintenance.

And now to sit down and build one

Here are a few pointers:

First and foremost, select the 3D printer you want to build. There is a range of options among open source 3D printers that you can choose from.

The best design to start out with is the Prusa i3. Named after Josef Prusa – the designer of i3 and its predecessor Prusa Mendel (the first printer I built!) – it's the machine of choice for many people and was the unofficial product of the RepRap project. There are several versions and upgrades available for it.

I had great success with the i3 model, as did many others whom I helped build it. The difficulty level of making a 3d printer is similar to assembling a computer, but the process is longer and